1. Literature Review Discussion The section reviews intelligent surveillance systems, noting a gap between general human action recognition research and its application to suicide prevention. It highlights that prior suicide detection systems only trigger alerts *after* an incident (such as jumps or hangings) occurs, rendering them ineffective for early prevention. It further explores studies identifying pre-suicidal behavioural indicators—such as pacing, leaving belongings behind, station hopping, or specific head poses—and reviews 2D skeleton-based action recognition techniques using static, motion, and geometric features processed via CNN, RNN, and LSTM architectures.

2. Methodology The authors propose a modular, top-down visual processing pipeline designed to handle pedestrian detection (using YOLOv5x), pedestrian tracking (using DeepSORT with motion/grouping features), and pose estimation (using HRNet). The core feature is a custom 2D skeleton-based action recognition model that extracts static, motion, and geometric features from body keypoints. These features are processed through a two-branch stacked LSTM network whose classification scores are merged via late multiply score fusion. A logical layer built on top applies domain knowledge from a human-coding study to detect single unit actions and complex pre-suicide behaviour patterns over time.

3. Limitations and How to Address Them

- **Small and Unbalanced CCTV Dataset:** Pre-suicide events are statistically rare, yielding limited real-world training samples and leading to poorer recognition for short actions like taking a photo or placing an object. *Fix:* Train with synthetic/acted recordings simulating target behaviours, supplement with routine footage, and employ weighted cross-entropy loss.
- **Frame-Wise Action Shifts and Tracking Errors:** Rapid shifts between similar postures (e.g., leaning vs. leaning head down) or ID-switching in tracking cause fragmented detection or missed complex patterns. *Fix:* Incorporate higher-level temporal grouping (action tubes), fine-tune thresholding logic, and enhance the tracking module for robust continuous trajectory analysis.
- **Camera Angle and Illumination Limitations:** Outdoor surveillance suffers from low resolution, distance degradation, and low performance during darkness. *Fix:* Retrain models when installing new camera angles, rely on viewpoint-invariant geometric features, and integrate thermal-based visual analysis for nighttime scenarios.

4. Method or Algorithm Used The paper utilizes a modular framework featuring **YOLOv5x** for pedestrian detection, **DeepSORT** for multi-object tracking, and **HRNet** for 2D human pose estimation. The key algorithmic contribution is a **2D skeleton-based action recognition algorithm featuring a dual-branch stacked LSTM neural network with late score fusion**, complemented by a **rule-based Logical Layer** that aggregates frame predictions into action tubes and tracks temporal risk patterns.

Deep Dive

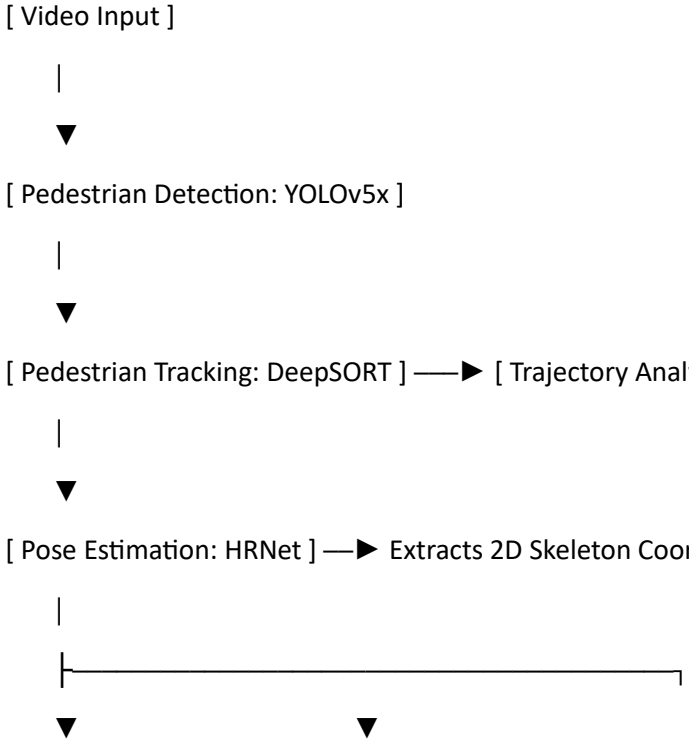
5. Advantages and Disadvantages of the Method

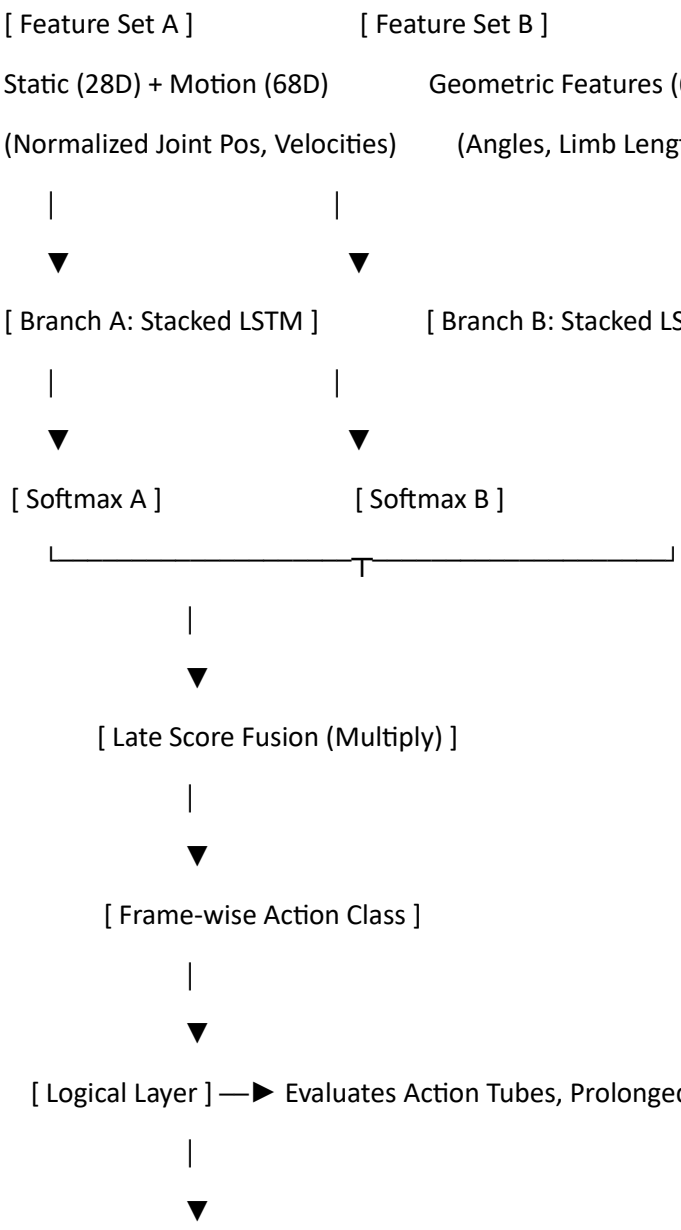
- **Advantages:**
    - *Privacy-Preserving & Lightweight:* Processing 2D skeleton joint coordinates rather than raw RGB frame volumes significantly reduces computational load while avoiding the storage of sensitive facial/identifying appearance data.

- *Robust to Visual Noise:* Skeleton keypoints are less affected by background clutter, lighting variations, and pedestrian clothing changes compared to standard optical flow or 3D-CNN methods.
- *Early Crisis Detection Capability:* Combines unit-action detection with a rule-based logical layer to capture complex pre-suicide behavioral patterns (e.g., pacing, crouch gestures, or prolonged leaning) prior to a suicide attempt.
- *Modular Pipeline:* Each component (detection, tracking, pose estimation, action recognition) operates independently, allowing modules to be swapped, updated, or re-used for other tasks like trajectory analysis.

- **Disadvantages:**
  - *Pipeline Error Propagation:* Because it uses a top-down modular approach, failure in early stages (such as a tracking ID swap or an inaccurate pose keypoint) directly compromises the final action classification.
  - *Difficulty with Subtle or Swift Actions:* The model struggles to differentiate brief, single-instance movements or actions with small joint position variations (such as placing an item or taking a photo).
  - *Dependency on Pose Estimation Quality:* Severely occluded pedestrians or low-resolution camera views lead to missing skeletal joints, degrading downstream classification performance.

6. Detailed Explanation of the Method: How It Works





1. **Pedestrian Detection & Tracking:** Frames pass through YOLOv5x to locate pedestrians. DeepSORT assigns unique IDs and calculates bounding-box velocity.
2. **Pose Estimation:** HRNet crops individual pedestrian regions and extracts 2D joint coordinates for 14 key body joints. Coordinates are normalized against body height using the neck point as the origin.
3. **Feature Extraction (per 30-frame window):**
    - *Branch A Features (96D):* Combines static normalized coordinates (28D) with motion features (68D)—including joint velocities, body velocity, and bounding-box velocity.

    - *Branch B Features (62D):* Extracts viewpoint-invariant geometric relationships: limb orientations, limb lengths, inner joint angles, long lengths, and end-joint distances.

4. **Dual-Branch Stacked LSTM Processing:** Each branch feeds its feature sequence through a fully connected layer, two stacked LSTM layers, a final dense layer, and a Softmax function to compute class probabilities.
5. **Score Fusion:** Frame predictions are computed by multiplying the probability score vectors from Branch A and Branch B element-wise.
6. **Logical Layer Processing:** Converts frame-level labels into continuous "action tubes". It applies domain rules (elicited from suicide prevention researchers) to detect:
    - Sensitive unit actions (*Crouch*, *LeanHead*, *Place*).
    - Prolonged states (e.g., sitting or leaning for >100 seconds).
    - Repetitive action patterns (e.g., shifting between sitting, walking, and standing repeatedly).

7. Suitable Contexts and Problem Types

- **Public Safety & Hotspot Monitoring:** Real-time automated surveillance at high-risk public infrastructure locations such as cliffs, bridges, railway platforms, and metro stations.
- **Early Crisis/Anomaly Prevention:** Scenarios requiring early detection of subtle behavioral cues prior to a critical event, rather than post-incident detection.
- **Resource-Constrained Edge Computing:** Environments running automated video analysis locally where raw video processing (like heavy 3D-CNNs) is computationally prohibited.
- **Privacy-Sensitive Surveillance:** Monitoring scenarios subject to strict privacy regulations where storing or processing raw RGB imagery of individuals is restricted.

8. Main Results and Findings

- **JHMDB Benchmark:** The model achieved **64.01%** accuracy on Split 1 (ground truth pose) and **60.2%** overall mean accuracy using its own automated processing pipeline, outperforming comparable skeleton models like PoTion and EHPI.
- **Dual-Branch Fusion Validation:** Combining static/motion features (Branch A: 62.81%) with geometric features (Branch B: 43.06%) via late multiply fusion significantly boosted performance to 64.01%, proving late fusion superior to early or mid-fusion strategies.
- **Real-World CCTV Dataset (Suicide Hotspot):**
    - *Action Classification:* Achieved an average accuracy of **82.12%** across 8 action classes. High performance was recorded for critical crisis actions like *LeanHead* (96.55% F1-score) and *Stand* (95.55% F1-score).

    - *Comparison to Baseline Models:* Outperformed traditional 3D RGB models (*SlowFast*) on fine-grained gestures like *Crouch* and *LeanHead*, while performing comparably to SOTA skeleton algorithms (*PoseC3D* at 84.39%) with a much lighter design.

- **Crisis Detection & Logical Analysis Performance:** On 15 simulated hotspot test clips, the pipeline achieved **86.7% accuracy** and **80% recall** at the clip/video level for identifying crisis situations, and a **72.7% precision** for capturing individual high-risk behavioral events.

9. Overall Opinion and Critical Evaluation

- **Strengths:**
    - **Interdisciplinary Innovation:** Successfully bridges computer vision with psychiatric/public health domain knowledge by translating qualitative human-coding insights into a rule-based logical layer.
    - **Pragmatic Architecture:** Selecting a modular skeleton-based LSTM rather than an end-to-end 3D-CNN ensures the system remains lightweight, interpretable, and adaptable to existing outdoor CCTV infrastructure.
    - **Solid Empirical Validation:** Thorough feature ablation experiments (comparing early, mid, and late fusion) justify the two-branch design.
- **Weaknesses:**
    - **Dataset Limitations:** The CCTV evaluation relies heavily on acted simulations (only 1 actor across 95 clips) rather than extensive real-world crisis footage, leaving its generalization across varied body types, clothes, and natural crowds unproven.
    - **Brittleness of Rule-Based Logic:** The logical layer relies on fixed hard thresholds (e.g., 100 seconds for prolonged state), which may lead to false positives/negatives in non-standard scenarios.
    - **Pipeline Cascade Failures:** Errors in bounding-box tracking directly cause tracking ID swaps, fragmenting temporal action tubes and leading to missed crisis warnings.
- **Verdict:** The paper represents a meaningful and pioneering effort in visual surveillance for early suicide prevention. While the pipeline demonstrates strong feasibility and competitive performance, real-world field deployment will require replacing the static rule-based logical layer with probabilistic temporal models and evaluating performance on broader, non-acted datasets.